Reviewer pnsQ

Subject

Niche Paper with model and framework not explained well enough

Overall assessment

- Overall positive review (Rating: 4 – Borderline Accept).
- Reviewer appreciates the novelty of modeling temporal dependence through the carry-over context and reward kernel.
- Likes the Gymnasium benchmark and the online learning formulation.
- Main concerns relate to clarity of presentation, the formal justification of the myopic-optimal valuation, and missing empirical analysis of its contribution and computational efficiency.

Major points to address

1. **Improve exposition of the mathematical framework**

   TODO: Reorganize Section 3.1 into clearer definitions instead of introducing all notation in one paragraph.

   TODO: Consider adding a notation table summarizing the main variables, symbols, and their meaning.

   TODO: Introduce the mathematical formulation more progressively before presenting the formal model.

   TODO: Use the Related Work section to motivate the formulation from classical repeated auction and bidding problems before introducing the new concepts.

2. **Formalize the notion of myopic-optimal valuation**

   TODO: Provide a precise definition of the myopic-optimal valuation.

   TODO: Clearly state the assumptions under which it is optimal (second-price auction with no future interaction).

   TODO: Explain why this approximation is expected to remain useful inside the MPC framework despite not being globally optimal.

   TODO: Strengthen the theoretical motivation connecting the approximation to planning performance.

3. **Ablation study for myopic-optimal valuation**

   We provide a formal ablation quantifying the planning gain attributable to multi-step horizon planning on top of the myopic-optimal terminal value. Results are presented in Tables **??** and **??** in the response draft below.

Requested experiments

1. **Performance versus computation**

   We report the full reward–latency trade-off for all MPC variants in Table **??** and Figure 1 of the response draft.

   We additionally report how reward evolves with the number of planning rollouts (M) in Table **??**, showing that reward saturates quickly with $M \geq 5$, establishing that MPC's planning gain stems primarily from the V_0 value head rather than rollout volume.

2. **Comparison with Deep RL**

   Table **??** reports one-time training cost (wall-clock hours on the same cluster), per-bid inference latency (ms), and mean episode reward for all policies, enabling a direct cost–benefit comparison.

Paper modifications

- **TODO:** Improve the readability of Section 3.1 with clearer structure and notation.
- **TODO:** Add a notation table summarizing symbols and variables.
- **TODO:** Introduce the myopic-optimal valuation more formally.
- **TODO:** Clarify why the approximation is theoretically justified within MPC.
- Include an ablation study for the myopic-optimal valuation (Tables **??**–**??**).
- Expand discussion of computational complexity and efficiency (Table **??**).

Response draft

We thank Reviewer pnsQ for the careful reading and constructive feedback. We address the theoretical points (Weaknesses 1–2) jointly with co-authors in a separate section of the rebuttal. Below we focus on the empirical questions: the myopic-optimal ablation (Weakness 3), the performance–computation trade-off (Question 1), and the RL vs. MPC computational comparison (Question 2).

All experiments below were performed on the `survival-second-price-exponential` (η=50) and `saturation-second-price-uniform` (η=50) benchmark scenarios included in our framework; the cluster is Margaret (`normal-best` partition, 4 CPUs per job).

W3 / Myopic-optimal ablation. *""A formal ablation study to quantify the improvement brought upon by myopic-valuation is missing.""*

The myopic-optimal policy, which we call FiPla, bids exactly the per-auction expected conversion reward $E[R_t \mid \text{context}_t]$. Under a second-price auction rule and conditioned on winning, this bid is exactly optimal when future auctions are ignored: it is the unique fixed point of the myopic Bellman equation. Any improvement of MPC over FiPla is therefore attributable *solely* to multi-step planning—FiPla sets the ceiling for all single-step strategies.

Table **??** reports the full planning-gain decomposition on the survival scenario (100 seeds, M=10 rollouts, timing benchmark conditions), and Table **??** provides the same analysis on the saturation scenario (1 000 seeds, full benchmark).

Table 1: Myopic-optimal ablation — survival scenario (`survival-second-price-exponential`, η=50, 100 seeds). Gain is computed relative to FiPla.

Policy	Planning component	Mean reward	Std	Gain vs. FiPla
FiPla	myopic-optimal (no planning)	1.606	1.591	—
GreedyMyopic	single-step, learned V_0	1.447	1.609	−9.9%
Shading(α=0.7)	single α, 10-rollout sim	1.673	1.866	+4.2%
MPC(H=10, K=1)	horizon H=10	1.958	1.742	+21.9%
MPC(H=25, K=1)	horizon H=25	2.227	2.151	+38.7%
MPC(H=10, K=2)	H=10 + time bins	1.996	1.758	+24.3%
MPC(H=25, K=2)	H=25 + time bins	2.278	2.255	+41.8%

Table 2: Planning gain over myopic-optimal — saturation scenario (`saturation-second-price-uniform`, η=50, 1 000 seeds). This scenario imposes a carry-over saturation effect that makes multi-step planning more valuable than in the survival scenario.

Policy	Mean reward	Std	Gain vs. FiPla
FiPla	1.392	1.857	—
PPO (tuned, ent=0.01)	1.288	1.616	−7.5%
SAC	1.030	1.946	−26.0%
MPC(H=10, K=1)	2.973	2.529	+113.6%
MPC(H=25, K=1)	2.860	2.486	+105.5%
MPC(H=10, K=2)	3.095	2.571	+122.3%
MPC(H=25, K=2)	3.032	2.560	+117.8%

Three findings emerge. First, FiPla strictly dominates all non-planning policies on the saturation scenario (+7.5% over the best RL baseline) and is competitive on survival, confirming its role as a meaningful oracle ceiling for single-step bidding. Second, every MPC variant beats FiPla by a substantial margin—between +22% and +42% on survival, and +106% to +122% on saturation—demonstrating that the planning horizon, not the terminal value approximation, drives MPC's advantage. Third, RL baselines (PPO, SAC, TD3) uniformly *underperform* FiPla on the saturation scenario despite orders-of-magnitude more training compute (see below), which shows that the bottleneck for deep RL is not sample efficiency but the difficulty of credit assignment in the presence of carry-over effects.

Number of rollouts (M) and V_0 dominance. Table **??** shows mean reward for MPC(H=10) as M varies from 1 to 50 (survival scenario, 30 seeds). Reward is essentially flat beyond M=5, with overlapping standard deviations at every point. This confirms that the planning gain measured in Tables **??**–**??** comes overwhelmingly from the MLP terminal-value head V_0, not from the rollout dynamics themselves. Increasing M only scales latency linearly without improving reward, which directly motivates the choice of a small default (M=10) in our benchmarks.

Table 3: Reward vs. number of planning rollouts M for MPC(H=10) (survival scenario, 30 seeds). Latency scales approximately linearly with M; reward is flat beyond M=5.

M (rollouts)	Mean reward	Std	Avg bid latency (ms)
1	1.655	1.575	≈ 9
5	2.082	2.267	≈ 47
10	1.727	1.665	≈ 94
25	1.735	1.703	≈ 234
50	1.768	1.633	≈ 469

Q1 / Performance–computation trade-off for MPC. *""What is the trade-off between computation and performance, when we replace full dynamic programming with myopic-optimal valuation?""*

Full dynamic programming is intractable in our setting (continuous auction arrival times, stochastic competition, heterogeneous user features, carry-over context). Our framework replaces it with a two-component approximation: (i) forward rollouts using FiPla bids—which are provably optimal at each step conditioned on the current carry-over context—to estimate future reward given an α-scaled bid today, and (ii) a learned V_0 MLP to estimate the campaign's terminal value. Table **??** reports the per-bid decision latency and mean reward for all four MPC variants used in our experiments.

Doubling the planning horizon from H=10 to H=25 costs 1.9× more latency. Adding time bins (K=2) multiplies latency by 6× without consistent additional reward at 100 seeds. In practice, H=25, K=1 offers the best reward–latency ratio.

The V_0 MLP pretraining that enables this decomposition requires only ≈ 5 minutes (≈ 0.08 h) on a single CPU node. This is several orders of magnitude cheaper than training deep RL policies (see Table **??**).

Q2 / Computational comparison: MPC vs. Deep RL. *""Computational load comparison between Deep RL techniques and MPC style approach.""*

Table 4: MPC variants: performance–cost trade-off (M=10 rollouts, Margaret `normal-best`, 4 CPUs). **All runs use 100 seeds: evaluating MPC(H=25, K=2) at 1 000 seeds would exceed 40 hours. Rewards are from these same 100-seed timing runs and therefore differ slightly from the 1 000-seed results reported elsewhere.** FiPla and Shading are shown as reference baselines.

Policy	Mean reward	Avg bid latency (ms)	Cost vs. cheapest MPC
FiPla	1.606	0.001	—
Shading(α=0.7)	1.673	24.1	—
MPC(H=10, K=1)	2.113	110.5	1.0×
MPC(H=25, K=1)	2.281	204.8	1.9×
MPC(H=10, K=2)	2.128	657.6	6.0×
MPC(H=25, K=2)	2.278	834.8	7.6×

Table **??** provides a comprehensive comparison of one-time training cost, per-bid inference latency, and mean episode reward for all policies. Training times for RL were measured from SLURM wall-clock records on the same cluster (10M environment steps, 6 scenario instances in parallel).

Table 5: Deep RL vs. MPC: training cost, inference latency, and reward (survival scenario, η=50; RL: 10M steps; MPC: V_0 pretraining only; latency measured on Margaret `normal-best`, 4 CPUs). **All runs use 100 seeds for fair comparison: evaluating MPC(H=25, K=2) at 1 000 seeds would exceed 40 hours. Rewards are from these same 100-seed timing runs and may differ slightly from the 1 000-seed results reported in Tables ?? and ??.** Performance ratio: $\eta = \bar{R}/\bar{R}_{\text{FiPla}}$ (η=1 matches FiPla; η>1 exceeds it; η<1 falls short). Cost metrics (lower is better): $c_{\text{train}} = t_{\text{train}}/\eta$ (training hours per unit η) and $c_{\text{infer}} = t_{\text{infer}}/\eta$ (ms per bid per unit η).

Policy	Type	Train. (h)	Infer. (ms)	η	c_{train} (h/η)	c_{infer} (ms/η)
FiPla	myopic	0.00	0.001	1.000	0.000	0.001
PPO	Deep RL	0.80	0.234	0.976	0.820	0.240
SAC	Deep RL	10.0	0.306	0.483	20.723	0.634
TD3	Deep RL	9.1	0.231	0.694	13.119	0.333
MPC(H=10, K=1)	MPC	0.08	110.500	1.316	0.061	83.986
MPC(H=25, K=1)	MPC	0.08	204.800	1.420	0.056	144.195
MPC(H=10, K=2)	MPC	0.08	657.600	1.325	0.060	496.290
MPC(H=25, K=2)	MPC	0.08	834.800	1.418	0.056	588.538

Several observations stand out.

- **Training cost**: MPC requires only V_0 pretraining ($\approx$ 5 min, 0.08 h), which is 10× cheaper than PPO (0.8 h) and 125× cheaper than SAC or TD3 (9–10 h).
- **Inference cost**: Deep RL performs a single forward pass at bid time (0.31 ms), whereas MPC must execute M=10 simulation rollouts per bid (111–835 ms depending on H and K). MPC is therefore 350–3 600× slower *per bid* than RL.
- **Reward**: No RL policy exceeds FiPla (1.606), while the cheapest MPC variant (MPC(H=10, K=1), 2.113) beats the best RL policy (PPO, 1.567) by +35%. The best MPC variant (MPC(H=25, K=1), 2.281) outperforms the best RL baseline by +45% at a fraction of the training cost.

- **Practical deployment**: Auction systems typically operate on 100 ms response-time budgets. MPC(H=10, K=1) at $\approx$ 110 ms is near this limit; larger variants require pre-computation or asynchronous planning.

In summary, MPC achieves substantially better reward at far lower training cost, at the price of higher per-bid latency. For latency-critical settings, H=10, K=1 provides the best reward–latency balance; for offline or asynchronous bidding, larger variants are preferable.

Cross-preset robustness. MPC consistently dominates across all four built-in presets, with the planning gain amplified in scenarios with stronger temporal dependence (recency: +267%, saturation: +114%).

Table 6: Cross-preset robustness (η=50, 1 000 seeds). Each cell shows mean reward (gain vs. FiPla). Shading uses the best α per scenario: 0.89 / 0.5 / 0.7 / 0.6 for Survival / Saturation / Recency / Mixed. Shading values updated from 200-seed sweeps to dedicated 1 000-seed runs.

Policy	Survival	Saturation	Recency	Mixed
FiPla	2.560 (—)	1.392 (—)	1.988 (—)	1.766 (—)
Shading (best α)	2.184 (−14.7%)	1.396 (+0.3%)	1.849 (−7.0%)	1.745 (−1.2%)
PPO	1.647 (−35.7%)	1.288 (−7.5%)	1.704 (−14.3%)	1.794 (+1.6%)
MPC(H=10)	4.008 (+56.6%)	2.973 (+113.5%)	7.293 (+266.8%)	3.126 (+77.0%)
MPC(H=10, K=2)	4.129 (+61.3%)	3.095 (+122.3%)	7.896 (+297.1%)	3.108 (+76.0%)

All PPO entries reflect a per-scenario hyperparameter search over 135 combinations (lr $\times$ ent_coef $\times$ n_steps $\times$ λ_{GAE}, 5×3×3×3 grid, 1 M steps per trial) followed by a full 10 M-step retrain with the best configuration. The updated Shading results (1 000 seeds) further narrow the gap with FiPla: Shading is at best marginally above FiPla on saturation (+0.3%) and below it on all other scenarios. MPC dominates all baselines by wide margins in every preset.